

Statistics for ML: four modules

1. Descriptive statistics

population and sample
types of data
central tendency
dispersion
quantiles and percentiles
skewness and kurtosis
univariate, bivariate and
multivariate analysis
covariance and correlation

2. Probability distributions

random variables
PMF, PDF and CDF
kernel density estimation
2D density plots
normal, uniform, Bernoulli,
binomial, log-normal, ...

4. Miscellaneous

Chebyshev's inequality
Q-Q plot
sampling techniques
resampling techniques
statistical moments
Bayesian statistics (intro)

3. Inferential statistics

central limit theorem
confidence intervals
hypothesis testing
z-test, t-test, chi-square
test, ANOVA
which test to use when

each module builds on the one before it;
the miscellaneous topics are studied when an algorithm needs them